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Quantitative Analysis of User Interaction and HCI Principles

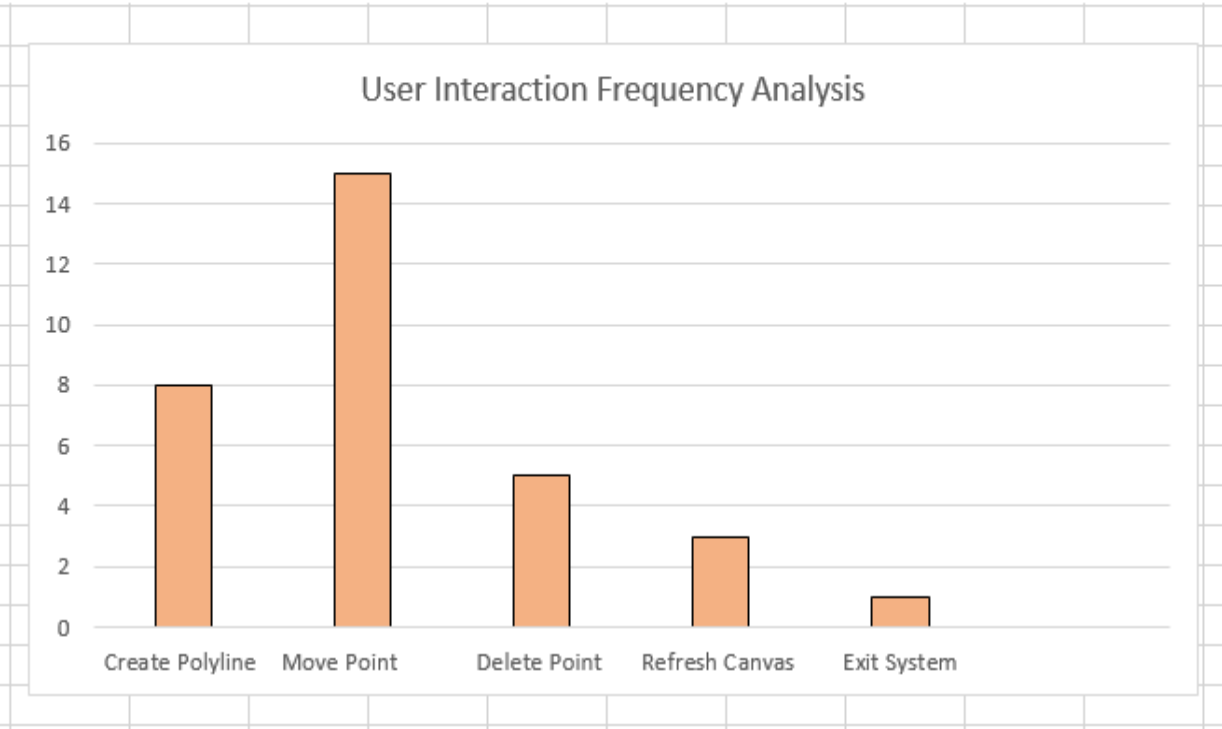
"This section presents a quantitative analysis of user interactions using graphical representations. The insights highlight key HCI principles such as usability, efficiency, and error handling."

Tool used : Microsoft Excel

1. "User Interaction Frequency Analysis"

The data used in this graph is based on estimated user interaction patterns, assuming typical usage of the Polyline Editor where moving points is the most frequent operation.

Action	Frequency
Create Polyline	8
Move Point	15
Delete Point	5
Refresh Canvas	3
Exit System	1

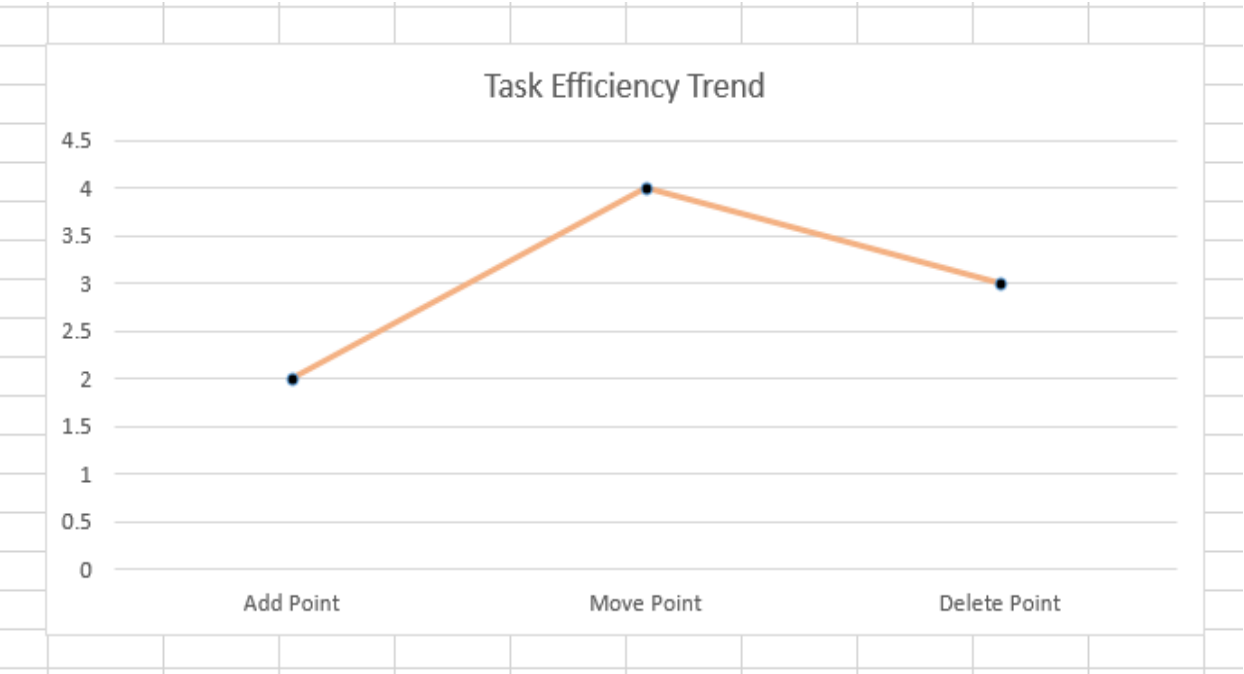


"This graph represents user interaction frequency, showing that the Move Point operation is the most frequently used, indicating the importance of efficient direct manipulation in the system."

2. "Task Time Comparison Graph"

The data used in this graph is hypothetical and based on assumed user interaction patterns. It represents an estimated time required to perform different tasks in the Polyline Editor, considering typical user behavior.

Task	Time (seconds)
Add Point	2
Move Point	4
Delete Point	3

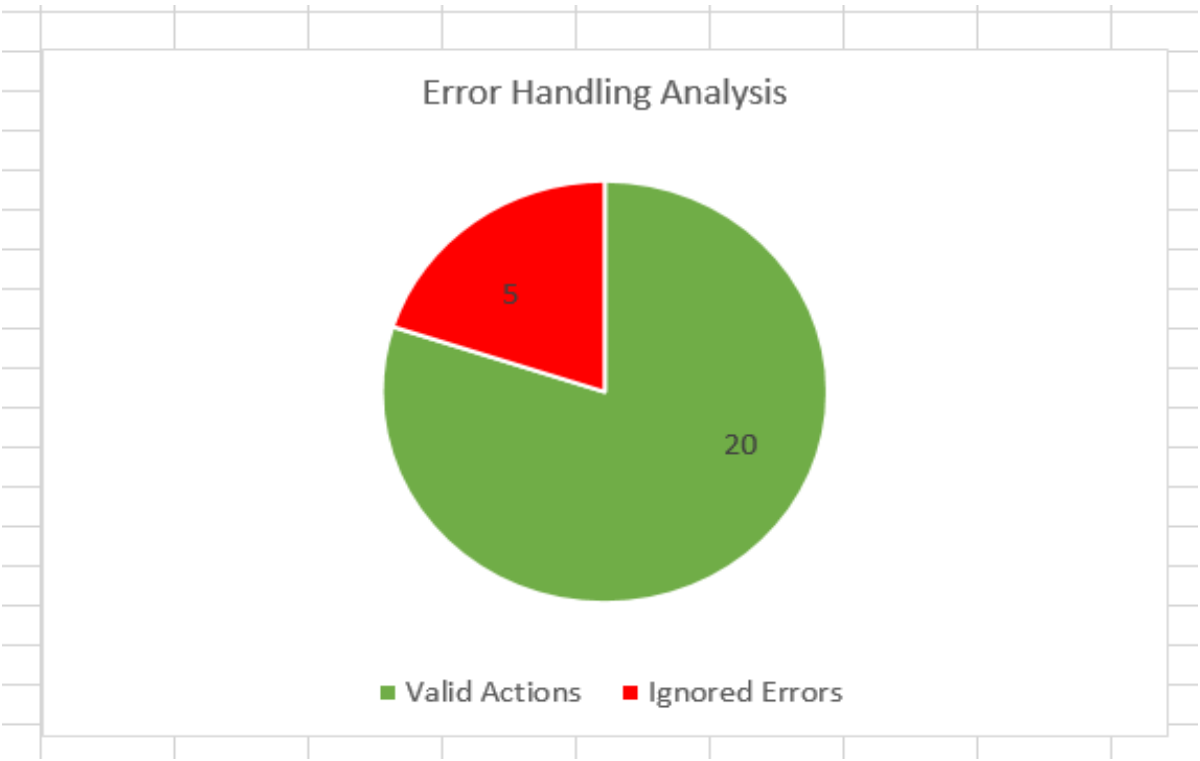


“Shows variation in time required for different tasks ,Move Point takes more time due to precision interaction .It indicates need for efficient direct manipulation design”

3. “Error vs Valid Actions Graph”

The data used in this graph is hypothetical and based on assumed user interactions. It represents a scenario where most user actions are valid, while a smaller portion includes ignored or invalid inputs.

Type	Count
Valid Actions	20
Ignored Errors	5



“Majority of actions are valid, indicating good system usability.
A small portion of inputs are ignored, showing effective error handling.
The system prevents incorrect operations, improving user experience and reliability.”

This analysis shows that the system is designed to handle errors efficiently while maintaining smooth user interaction.